On this page

Monitoring Guide

Case Manager uses LetMeKnow library to export different application metrics.

The metrics are available in the following HTTP endpoints:

URI	Description
`/api/metrics`	The metrics in the dropwizard format.
`/api/metrics/rich`	The metrics in the LetMeKnow format.
`/api/metrics/prometheus`	The metrics in the Prometheus format.
`/api/healthchecks`	CM healthchecks in CSV format.

Health Checksâ□□

CM is exporting the following health checks to check if the following dependencies are working properly:

- Database
- Messaging Server
- Pulse
- Tokenizer
- Event Ingestion

The Health Checks catch all the exceptions thrown and log them. If any of the healthchecks fail, please refer to the CM logs for more details on the error.

Database Health Checkâ□□

The database health check relies on the shared CM's connection pool to execute the statement SELECT 1. If the connection is successfully retrieved and the statement finishes, the health check succeeds, if any error occurs in any of the steps the check fails. A timeout of 5 seconds was given when retrieving a connection to avoid hanging the health check response.

Messaging Health Checkâ□□

The Messaging Server Health Check is achieved by trying to initialize a new connection with the messaging server.

Tokenizer Health Checkâ□□

The tokenizer health checks validates if there is a tokenizer instance reachable and if the authorization configured is valid.

Pulse Health Checkâ□□

The Pulse Health Check when triggered tries to connect to Pulse and tries to login, if it succeeds the check will also succeed.

Event Ingestion Health Checkâ□□

The Event Health Check checks if the current Case Manager node is taking events from the event RabbitMQ, or if it is blocking this action.

Metricsâ

CM already exports the default metrics provided by [LetMeKnow](). On top of these metrics, CM also exports metrics related to the following components:

Connection Poolâ

Metric	Description	Tags	Type
`cm.db.connection.pool.size`	The current size of the connection pool (If pool is lazy the value can grow over time)	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.used`	The number of used database connections.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.free`	The number of free database connections.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.waiting`	The number of threads waiting for a database connection	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.usage`	The current usage of the database connection pool (used/size in percent)	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

Event Ingestionâ

The latency metric has 3 dimensions specified by the tag "eventType" each dimension refers to the type of the event the metric is measuring.

Metric	Description	Tags	Type
`cm.event.processing.deadletter.count`	The total number of dead letters since CM's startup.	appType=CASE_MANAGER metricType=APPLICATION	Counter
`cm.event.processing.latency`	The event processing latency and rate aggregated over a 10 minute sliding window.	appType=CASE_MANAGER metricType=APPLICATION eventType={NEW_EVENT, STATUS_CHANGE, NEW_EXPLANATION}	Timer
`cm.event.processing.queue.used.count`	The number of events waiting to be processed.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.event.processing.queue.ratio`	The percentage of the event processing queue being used.	appType=CASE_MANAGER metricType=APPLICATION	Gauge

CM APIâ

The metrics below can have several dimensions defined by the tag "path" which indicates the URI of the API the metric is measuring.

Metric	Description	Tags	Type
`cm.request.api`	The latency of the endpoint described in the path tag, aggregated in a 10 minutes bucket.	appType=CASE_MANAGER metricType=APPLICATION	Timer

Metric	Description	Tags	Type
		path={ The API path the metrics refers to}	
`cm.request.api.error`	The rate of errors returned by the endpoint described in the path tag.	appType=CASE_MANAGER metricType=APPLICATION path={ The API path the metrics refers to}	Meter

Tokenizer / Encryptionâ

The metrics below can have two dimensions defined by the tag "operation" which indicates the operation the metric is measuring.Â

Metric	Description	Tags	Type
`cm.request.tokenizer`	The latency of the Tokenizer operation described, aggregated in a 10 minutes bucket.	appType=CASE_MANAGER metricType=APPLICATION operation={ The operation the metrics refers to (getToken, getPlainText, getBulkToken, getBulkPlainText, encryptData)}	Timer
`cm.request.tokenizer.error`	The number of errors returned by the Tokenizer operation described.	appType=CASE_MANAGER metricType=APPLICATION operation={ The operation the metrics refers to (getToken, getPlainText, getBulkToken, getBulkPlainText, encryptData)}	Gauge
`cm.request.tokenizer.error.percent`	The percentage of errors returned by the Tokenizer operation described.	appType=CASE_MANAGER metricType=APPLICATION operation={ The operation the metrics refers to (getToken, getPlainText, getBulkToken, getBulkPlainText, encryptData)}	PercentGauge

Pulse Requestsâ

The metrics below can have several dimensions defined by the tag "path" which indicates the URI of the API the metric is measuring.

Metric	Description	Tags	Type
`cm.request.pulse.api`	The latency of the Pulse endpoint described in the path tag, aggregated in a 10 minutes bucket.	appType=CASE_MANAGER metricType=APPLICATION path={ The API path the metrics refers to}	Timer
`cm.request.pulse.api.error`	The rate of errors returned by the Pulse endpoint described in the path tag.	appType=CASE_MANAGER metricType=APPLICATION path={ The API path the metrics refers to}	Meter

Automation Rulesâ

The metrics below can have several dimensions defined by the tag "ruleName" which indicates the Automation Rule the metric is measuring.

Metric	Description	Tags	Type
`cm.automation`	The latency of the automation execution.	appType=CASE_MANAGER metricType=APPLICATION ruleName={ The rule name the metrics refers to}	Timer
`cm.automation.error`	The rate of errors returned by the automation execution.	appType=CASE_MANAGER metricType=APPLICATION ruleName={ The rule name the metrics refers to}	Meter

Loginâ

`cm.request.login`	The number of login requests.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.request.login.2fa`	The number of 2fa challenges requested during login requests.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.request.login.error`	The number of login requests with errors.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.request.login.error.percent`	The current percentage of login requests with errors.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

CM's Worker Threads - Event Ingestionâ

`cm.executor.pool.activeThreadsCount`	The total number of active threads in the executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.pool.taskQueueEventsCount`	The total number of tasks submitted to the executor and pending for execution.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.pool.maxPoolSize`	The current size of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.pool.threadsUsageRatio`	The current usage of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

CM's Worker Threads - Automation Rules Processingâ

`cm.executor.automations.pool.activeThreadsCount`	The total number of active threads in the executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.automations.pool.taskQueueEventsCount`	The total number of tasks submitted to the executor and pending for execution.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.automations.pool.maxPoolSize`	The current size of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.automations.pool.threadsUsageRatio`	The current usage of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

CM's Batchesâ

Case Manager uses several batches to persist data in the database. The following metrics measure the batch's queue occupancy percentage and the flush latency.

`cm.event.processing.queue.occupancy.[batch-name]`	The occupancy percentage of batch's queue. This queue is used as a buffer before sending events to the batch.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge
`cm.event.processing.batch.flush.latency.[batch-name]`	The latency to flush a batch to the database in entries/s.	appType=CASE_MANAGER metricType=APPLICATION	Timer

Dead Letter Metricsâ

Learn more on the dead letter metrics in the Dead Letters System page.

Async Extensionsâ

The following metrics can have several dimensions defined by the tag "extension" which indicates the name of the Extension the metric is measuring.

Metric	Description	Tags	Type
`cm.extension`	The latency of the async extension execution.	appType=CASE_MANAGER metricType=APPLICATION extension={ The extension name the metrics refers to}	Timer
`cm.extension.error`	The rate of errors returned by the extensions execution.	appType=CASE_MANAGER metricType=APPLICATION extension={ The extension name the metrics refers to}	Meter

Learn more on Monitoring Extensibility.

Webhook Metricsâ

The following metrics can have several dimensions defined by the tag URL which indicates the webhook URL that the metric is measuring.

Metric	Description	Tags	Type
`cm.webhook.total`	The total of executed webhook requests.	appType=CASE_MANAGER metricType=APPLICATION url={ The webhook URL }	Counter
`cm.webhook.error`	The total of failed webhook requests.	appType=CASE_MANAGER metricType=APPLICATION url={ The webhook URL }	Counter